

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DIGITAL SYSTEMS AND COMPUTER ORGANIZATION [CSS 2104]
COMPUTER SCIENCE AND ENGINEERING
SET A: ENSEMBLE OF PAST-PAPER QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-4; module 5 through memory operations. Independent practice paper; not an official university examination.

Section A: Multiple-choice questions (5 marks)

- 1) What is the range of 5-bit two's-complement integers? (1)
(A) -15 to 15
(B) -16 to 15
(C) -16 to 16
(D) 0 to 31
- 2) What is the 5-bit two's-complement representation of -9 ? (1)
(A) 01001
(B) 11001
(C) 10110
(D) 10111
- 3) Which addition overflows the 5-bit two's-complement range? (1)
(A) $-9 + 5$
(B) $6 - 13$
(C) $11 + 7$
(D) $-4 + 3$
- 4) Which arrays are programmable in a programmable logic array (PLA)? (1)
(A) Both AND and OR
(B) Only AND
(C) Only OR
(D) Neither
- 5) A PLA implements $F_1 = AB + AC$ and $F_2 = AB + BC$. Which product term can be shared? (1)
(A) AC
(B) AB
(C) BC
(D) ABC

Section B: Theory questions (25 marks)

- 6) Simplify $F(A, B, C, D) = \Pi M(0, 3, 5, 6, 8, 10)$, with don't-care terms $d(2, 13, 14)$, and implement it using NOR gates only. Take A as the most significant bit. (3)
- 7) Design a 3-bit magnitude comparator for unsigned binary numbers $A = a_2a_1a_0$ and $B = b_2b_1b_0$. Derive the Boolean expressions for $A > B$, $A = B$ and $A < B$, and draw the equivalent logic circuit. (3)
- 8) Implement a three-input XOR gate using 4-to-1 multiplexer with a neat diagram. (3)
- 9) Design a sequential circuit with two JK flip-flops A and B and two inputs E and F. If $E = 0$, the circuit remains in the same state regardless of the value of F. When $E = 1$ and $F = 1$, the circuit goes through the state transitions from 00 to 01, to 10, to 11, back to 00, and repeats. When $E = 1$ and $F = 0$, the circuit goes through the state transitions from 00 to 11, to 10, to 01, back to 00, and repeats. (5)
- 10) Design a 3-bit synchronous binary up-counter using T flip-flops. Draw the logic diagram. (3)
- 11) Illustrate a 4-bit asynchronous ripple counter with a neat diagram using JK Flip flops and explain the working of the same. (3)
- 12) Write behavioral Verilog code that represents an eight-bit ring counter. (2)
- 13) Explain the concept of memory addresses in an Instruction Set Architecture. Also, discuss the difference between byte addressing and word addressing. (3)

End of question paper